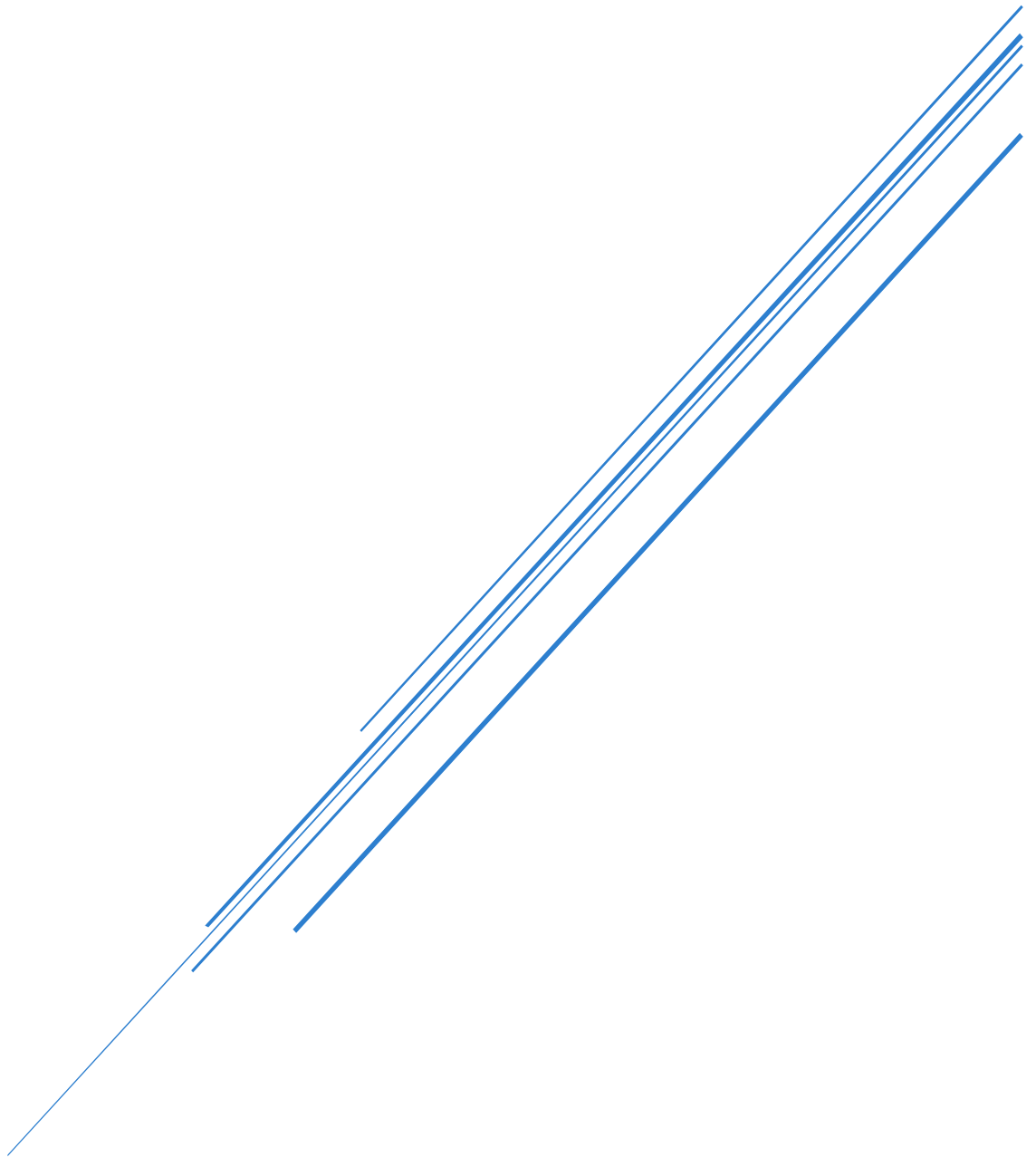


PROJECT PROPOSAL

Individual Challenge - Dutch Road Sign Recognizer



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[Course title]

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1. Introduction

Preparing for the Dutch CBR theory exam involves memorizing a large and varied set of road signs. For a person getting their driver's license in the Netherlands who is unfamiliar with Dutch traffic signage, understanding the meaning of an unfamiliar sign encountered in a practice question maybe be an issue. Practice applications such as the official CBR app and theorie.nl display signs with Dutch-language explanations, which may be difficult to interpret without Dutch language proficiency or prior driving experience in the Netherlands.

This project addresses that friction. The goal is to build a CNN image classifier that predicts the type of a Dutch road sign from a phone-camera photo, and returns the official Dutch name of the sign alongside a plain-English explanation of the rule it enforces. The output is a concrete, actionable identification that a theory-exam student can use in the moment, rather than having to look up the sign manually or ask a Dutch speaker after the fact.

The stakeholder for this project is a person currently getting their driver's license in the Netherlands and actively preparing for the CBR theory exam. Their experience with unfamiliar signs during practice shapes which sign classes to prioritize, what level of accuracy is considered useful, and what a meaningful output looks like from the perspective of an end user sitting a theory exam.

The final deliverable is a trained CNN classifier with architecture comparison output, demonstrating which backbone and training configuration gives the best accuracy on Dutch road signs from both a standard dataset and real-world photos collected in the Netherlands.

2. Timeline

The project follows the AMD Project Methodology and is divided into four phases.

Phase 1 - Problem Exploration (Completed)

Project setup and domain exploration. Challenge topic selected after scope narrowing from a multi-classifier architecture to a single road-sign classifier, based on coach feedback.

Phase 2 - Data Exploration and Stakeholder Alignment (In progress)

Mapillary Traffic Sign Dataset downloaded and explored. NL-relevant class subset being identified from the 313 available classes.

Phase 3 - Modelling and First Iteration (Upcoming)

Proposal finalized. NL-class filter applied. Baseline pretrained classifier trained on full Mapillary set to validate the pipeline, then retrained on the NL subset. Field-photo collection in the Netherlands planned for week 4.

Phase 4 - Evaluation, Deployment, and Submission (Upcoming)

Architecture comparison completed. Failure-mode analysis documented. Flask and React demo built. Stakeholder usability sessions conducted. Final model selected. Portfolio evidence compiled.

3. Domain Understanding

3.1 The Challenge of Learning Dutch Road Signs

The Dutch road sign system contains a large number of distinct sign types, many of which differ from signs used in other countries and are not immediately interpretable from their visual design alone. For a person preparing for the Dutch CBR theory exam who has not previously driven in the Netherlands, encountering an unfamiliar sign in a practice question causes confusion: the sign is displayed, but its meaning is not self-evident, and the Dutch-language explanation provided by the practice app may not be accessible to someone without Dutch language skills.

Theory-exam practice sessions surface many different sign types across many questions, and the same unfamiliar sign may appear repeatedly before its meaning is fully understood. The result is that exam preparation is slowed down.

3.2 Existing Tools and Their Limitations

The most direct existing resource is the official CBR practice app and its equivalents such as theorie.nl. These tools display signs with Dutch-language descriptions and link each sign to the relevant traffic rule. However, they do not offer an image-recognition interface: the user must already know which sign they are looking at in order to look it up. If the sign appears in a practice question, the user must either read the Dutch explanation directly, search for the sign manually using a description, or ask someone else.

General image-search tools can return the name of a sign from a photo, but they do not consistently provide the official Dutch name, the associated traffic rule in plain English, or a confidence measure that indicates whether the identification is reliable.

3.3 The Opportunity

A CNN classifier trained on labelled road-sign images can address both of these limitations. Given a photo of a sign, the model returns the predicted sign class, its official Dutch name, and a one-line plain-English explanation of the rule it enforces.

The feature that distinguishes this from a general image search is specificity: the classifier is trained on a dataset of road-sign images, filtered to the classes that appear in Dutch signs, and evaluated on photos collected.

4. Analytic Approach

4.1 Research Question

Can a fine-tuned CNN classifier, trained on the NL-filtered subset of the Mapillary Traffic Sign Dataset, recognize Dutch road signs from a phone-camera photo accurately enough to help a person studying for the Dutch CBR theory exam identify signs they encounter in practice questions?

Four sub-questions frame the iteration loop:

1. Which subset of Mapillary's 313 traffic-sign classes appears in Dutch signage, and how many training examples does each NL-relevant class have after filtering?
2. How much accuracy is lost when the model is evaluated on photos collected in the Netherlands versus the Mapillary held-out test split, and where does the model fail most?
3. Does the predicted class, Dutch name, and plain-English explanation actually reduce the stakeholder's confusion during theory-exam practice?

4.2 What Kind of Problem Is This?

This is an image classification problem. The target variable is a categorical label representing the road-sign class present in the input photo.

The input is a single phone-camera photo containing one road sign as the dominant subject. The output is a probability distribution over the NL-relevant sign classes. This satisfies the challenge requirement that the final product predicts something, and the prediction has a direct use: identifying an unknown sign during theory-exam practice.

4.3 What Steps Need to Be Taken?

4.3.1 Dataset Filtering

The Mapillary Traffic Sign Dataset covers 313 sign classes worldwide. The first step is to identify which of those classes correspond to signs that appear in Dutch road signage.

4.3.2 Data Preparation

The filtered dataset will be split into training, validation, and held-out test sets following standard practice.

4.3.3 Architecture Comparison

Multiple pretrained backbones will be compared, results will be compared on both the Mapillary held-out test split and the field-collected Netherlands evaluation set. The architecture comparison is the AMD-cycle evidence for the Modelling step.

4.3.4 Failure-Mode Analysis

After the architecture comparison, per-class accuracy will be computed. The bottom-performing sign classes and the most common confusion pairs will be identified and explained. This analysis drives the targeted augmentation applied in iteration 3 and documents where the model fails and why.

4.3.5 Evaluation Against Stakeholder Threshold

The model's practical usefulness will be evaluated against a success threshold defined with the stakeholder: whether the correct sign class appears in the top-1 or top-3 predictions often enough to be useful during theory-exam practice.

5. Data Requirements

5.1 Objective

The model requires labelled images of road signs that cover the classes present in Dutch road signage, with enough examples per class to support fine-tuning a pretrained backbone. A supplemental real-world evaluation set is required to measure accuracy under the conditions the stakeholder will actually use the app.

5.2 Data Type and Characteristics

The primary data is image data: JPEG and PNG photos of road signs with class-level annotations and bounding-box coordinates.

5.3 Data Sources

Dataset 1: Mapillary Traffic Sign Dataset (primary training source)

- Source: <https://www.mapillary.com/dataset/traffic-sign>
- Licence: Mapillary research licence (non-commercial academic use). Verifiable primary source. Not Kaggle.
- Volume: approximately 52,000 images, 313 sign classes worldwide, with bounding-box and class annotations.
- Reference: Ertler et al., Mapillary Traffic Sign Dataset for Detection and Classification on a Global Scale, ECCV 2020.

5.4 Data Legality and Ethics

The Mapillary Traffic Sign Dataset is published under the Mapillary research license for non-commercial academic use. It contains no personally identifiable information.

5.5 Data Diversity and Volume

The Mapillary dataset covers signs from multiple countries and road environments, providing variety in lighting, angle, weather, and sign condition. After NL-class filtering, the per-class sample count will be verified before training; classes with fewer than a minimum threshold of examples will be flagged, and the scope or augmentation strategy adjusted accordingly.

6. Requirements

Requirements are prioritized using MoSCoW, requirements are expressed at the business, functional, and non-functional level.

ID	Requirement	MoSCoW
BR-01	Develop a CNN image classifier that predicts the Dutch road sign class present in a photo and returns the official Dutch sign name and a plain-English explanation of the rule it enforces.	Must
FR-01	The model must accept a single phone-camera photo as input.	Must
FR-02	The model must output a predicted sign class from the NL-filtered Mapillary taxonomy.	Must

ID	Requirement	MoSCoW
FR-03	The model must output the top-3 predicted classes with confidence scores alongside the top-1 result.	Must
FR-04	The model must be trained and evaluated on real labelled image data, not synthetic data.	Must
FR-05	The model must output per-class accuracy.	Should
FR-06	A minimal Flask and React demo must accept an uploaded photo and return the predicted class, Dutch name, and plain-English explanation.	Should
NF-01	Top-1 accuracy on the Mapillary NL-filtered held-out test split should fall in the 85 to 95% range after the final iteration.	Should
NF-02	The project must follow the AMD Project Methodology throughout.	Must
NF-03	The data source must be a verifiable source, not Kaggle.	Must
NF-04	The domain and analytic approach must differ from the practice challenge.	Must

7. Sources

1. Mapillary, Traffic Sign Dataset: <https://www.mapillary.com/dataset/trafficsign>
2. Ertler et al. (2020). Mapillary Traffic Sign Dataset for Detection and Classification on a Global Scale. ECCV 2020.
3. CBR, Official Theory Exam Practice: <https://www.cbr.nl>